

Project Report

Auto Ticket Classification using Flow Designer

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1. INTRODUCTION

1.1 Project Overview

The school IT helpdesk handles daily incident requests from students and teachers, such as Wi-Fi problems, projector failures, forgotten passwords and slow computers. Today an IT agent reads every ticket and picks the category and subcategory manually. This project uses ServiceNow Flow Designer to classify each incident automatically when it is created, by checking keywords in the Short Description, and then to send the caller a confirmation email.

1.2 Purpose

- Classify incidents automatically at creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Deliver a no-code solution that is easy to maintain.

2. IDEATION PHASE

2.1 Problem Statement

IT staff manually review every incident to identify the issue type, choose the category and subcategory, and tell the caller that a ticket exists. This is slow, error-prone and does not scale as the number of requests grows.

Problem Statement	I am	I am trying to	But	Because	Which makes me feel
PS-1	an IT helpdesk agent at a school	handle student and teacher tickets quickly	I must classify every ticket by hand	there is no automatic categorization	overloaded and slow to respond
PS-2	a student or teacher	report an IT issue and get confirmation	I never hear whether the ticket was received	notifications are sent manually	uncertain and frustrated

2.2 Empathy Map Canvas

	Student / Teacher (Caller)	IT Helpdesk Agent
Says	"My WiFi is not working." "Did you get my ticket?"	"Which category is this?" "Too many tickets today."
Thinks	Will anyone fix this quickly?	Sorting tickets wastes my time.
Does	Types a short issue and submits it	Reads each ticket and selects category and subcategory
Feels	Anxious if there is no acknowledgement	Pressured, worried about wrong categories

2.3 Brainstorming

- Idea 1: Abusinessrule with a script to match keywords (rejected, needs scripting).
- Idea 2: Machine learning / Predictive Intelligence (rejected for now, needs training data).
- **Idea 3 (selected): Flow Designer with If / Else If keyword conditions, no code, easy to change.**
- Idea 4: Dependent Category/Subcategory choices for clean data.
- Idea 5: Automatic confirmation email to the caller.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Caller action	System response	Outcome
Report	Opens the form, enters Caller and Short Description	Auto-number assigned (INC prefix)	Ticket created
Classify	None	Flow reads Short Description and sets Category and Subcategory	Ticket correctly categorized
Acknowledge	Checks mailbox	Email: Your Request for the issue has been submitted	Caller informed
Resolve	Waits	Agent works the ticket by category	Faster routing

3.2 Solution Requirement

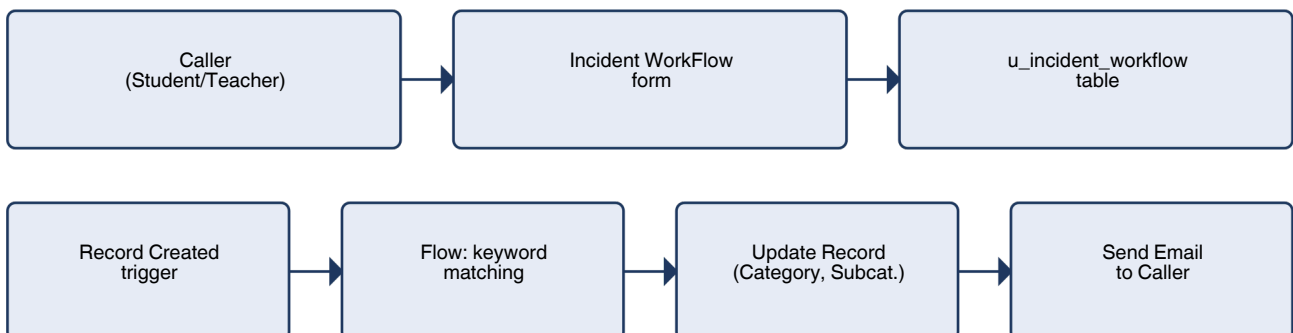
Functional requirements

- Automatically classify tickets from the issue description.
- Assign both Category and Subcategory without manual work.
- Support dependent choices between Category and Subcategory.
- Email the caller when a ticket is created.
- Store ticket data in a structured, standard format.

Non-functional requirements

- Easy to maintain and extend (no code).
- Secure: access restricted by roles and ACLs.
- Reliable: classification and email happen on every new ticket whose Category is empty.

3.3 Data Flow Diagram



Level 1 data flow: the caller's input is stored in the table, the trigger starts the flow, the flow updates the same record and emails the caller.

3.4 Technology Stack

Layer	Technology
Platform	ServiceNow (Global scope)
Automation	Flow Designer / Workflow Studio
Data	Custom table u_incident_workflow, Choice and Reference fields
Notification	Send Email action (SMTP)
Deployment	Local Update Set exported as XML

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution fit
Manual categorization is slow and inconsistent	Keyword rules in a flow classify every ticket the same way
Callers are not told a ticket exists	Flow sends an automatic confirmation email
Free-typed categories give messy data	Dependent choice lists enforce valid Category/Subcategory pairs
Needs frequent change by non-developers	No-code flow can be edited visually

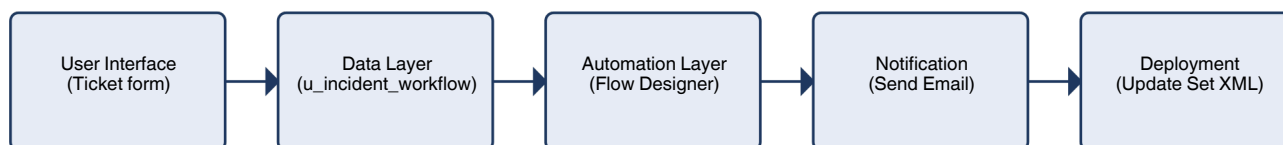
4.2 Proposed Solution

The caller enters only Caller and Short Description. A flow on the Incident WorkFlow table runs when a record is created with an empty Category. It checks the Short Description for keywords and applies these rules:

Keyword in Short Description	Category	Subcategory
Wi-Fi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot password	Access	Forgot Password
Slow Computer	Performance	Slow Computer

After classification the flow sends the caller an email titled Your Request for the issue has been submitted.

4.3 Solution Architecture



Layered architecture of the solution on the ServiceNow platform.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Deliverable
1. Requirement Analysis & Planning	Business use case, requirements, create Project Update Set	Requirements and update set
2. Backend Development & Configuration	Create table, auto-number, fields, choices, Category/Subcategory dependency	Configured Incident WorkFlow table

Phase	Activities	Deliverable
3. Automation	Build flow: trigger, If/Else If rules, Update Record actions, Send Email, Activate	Active flow
4. Testing, Validation & Security	Data validation checks, test scenarios, email check, role and ACL review	Test results
5. Deployment & Documentation	Set update set to Complete, Export to XML, final documentation	XML update set and report

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test	Input	Expected	Result
TS-01	WiFi not working in library	Network / Wi-Fi; email sent	Pass
TS-02	Projector not turning on	Hardware / Projector; email sent	Pass
TS-03	Forgot password	Access / Forgot Password; email sent	Pass
TS-04	Slow computer	Performance / Slow Computer; email sent	Pass
DV-01	Mandatory fields	Caller and Short Description required	Pass
DV-02	Auto-number	Unique INC numbers, no duplicates	Pass
DV-03	Reference fields	Resolve to sys_user and sys_user_group	Pass

Classification and the email were observed right after the record was saved and reloaded; no manual step was needed.

7. RESULTS

7.1 Output Screenshots

The screenshot displays the ServiceNow 'Dictionary Entry - Subcategory' page in 'Advanced' view. It shows a table of choices with the following data:

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Forgot Password	forgot password	en	2	false	2026-09-28 02:41:25	Access
Wi-Fi	wi-fi	en	0	false	2026-09-28 06:51:48	Network
Slow Computer	slow computer	en	3	false	2026-09-28 06:52:29	Performance
Projector	projector	en	1	false	2026-09-28 06:52:01	Hardware

The interface includes navigation tabs for 'Access Controls', 'Choices (4)', 'Attributes (1)', and 'Labels (1)'. The 'Choices' tab is active, showing the table above. The 'Dependent value' column is highlighted in green for the 'Forgot Password' row.

Subcategory choices with dependent values (Network, Hardware, Access, Performance)

Incident Workflow
INC00511

Number: INC00511

Caller: Abraham Lincoln

Category: Network

Subcategory: Wi-Fi

Short Description: Network issue

Description:

State: -- None --

Assigned Group:

Assigned to:

Update Delete

INC00511: Category Network, Subcategory Wi-Fi assigned automatically

Incident Workflow
INC00512

Number: INC00512

Caller: Amos Linnan

Category: Hardware

Subcategory: Projector

Short Description: Projector not turning on

Description:

State: -- None --

Assigned Group:

Assigned to:

Update Delete

INC00512: Category Hardware, Subcategory Projector assigned automatically

2026-09-28 10:01:35	admin@example.com	[ITSM Analytics] Daily Data Collection completed with error	send-ready	SMTP	(empty)
2026-09-28 00:00:21	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
2026-09-28 01:00:10	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-09-28 01:00:10	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-09-28 01:00:09	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-09-28 23:28:48	abraham.lincoln@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)

Confirmation emails recorded in System Logs, Emails

Preview Email

Ticket confirmation message.

Ref:MSG0000172_49PX6Q6aV7aWsdT-v9jEx

Close

Update Delete

Related Links
Preview Email

Email Log Email Attachments

Created Search

Email Log

Q	Created	Level	Message	Source
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Preview of the confirmation email

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Consistent classification and fewer human errors	Keyword matching misses tickets that use different wording
Less manual effort for IT agents	Spelling variants (WiFi / Wi-Fi) must each be listed
Callers get instant acknowledgement	Only four issue types are covered

Advantages	Disadvantages
No code; easy to maintain and extend	Does not yet assign group or priority
Portable through an update set	Email depends on the instance's outbound email setup

9. CONCLUSION

The project delivers an end-to-end, no-code automation for a school IT helpdesk. Tickets are classified automatically from the Short Description, dependent choices keep the data clean, and callers receive an instant email. Testing confirmed correct classification and email delivery, and the work was packaged as an update set. The approach suits small and medium environments such as schools.

10. FUTURE SCOPE

- Automatic assignment to the right group and agent.
- SLA tracking and priority rules.
- Predictive Intelligence for classification beyond fixed keywords.
- Keyword lookup table to add new issue types easily.
- Resolution and status-change emails to the caller.

11. APPENDIX

Item	Details
Source Code	Not applicable. The solution is built with Flow Designer (no code). Export: Project Update Set (XML).
Dataset Link	Not applicable. Data is stored in the table u_incident_workflow.
GitHub & Project Demo Link	[Add your GitHub / demo link here]